

DTS 304 – Data Management I

Tutor-Marked Assessment Report

Title: Design and Implementation of a Scalable Information Management System for a Retail Business

Name: Akhigbe Peace

Matric Number: 2024/A/SENG/0111

Student ID: 30107088

1. Introduction

The aim of this project was to design and partially implement an information management system for a retail business. The project involved collecting customer purchase data, storing it in a relational database, creating indexes to improve data retrieval, implementing SQL queries for data retrieval, and comparing sequential search with binary search using Python.

2. Data Capture and Representation

A Google Form was created to collect customer purchase information. The form captured the following fields:

- Customer Name
- Purchase Date
- Product ID
- Product Category
- Amount

A total of **30 customer purchase records** were collected and exported as a CSV file for further processing.

3. Database Design and Implementation

A MySQL database named **retail_business** was created to store the collected information.

A relational table named **purchases** was created with the following fields:

Column	Data Type
customer_id	INT (AUTO_INCREMENT, PRIMARY KEY)
customer_name	VARCHAR(100)
purchase_date	DATE
product_id	VARCHAR(20)
product_category	VARCHAR(50)
amount	DECIMAL(10,2)

The customer purchase data was successfully imported from the CSV file into the database. During the import process, the purchase date values were converted into MySQL's standard DATE format using SQL date conversion functions.

4. Indexing

To improve retrieval performance, the following indexes were implemented:

- Primary Index on **customer_id**
- Secondary Index on **product_category**
- Non-clustered Index on **purchase_date**

These indexes improve search efficiency by reducing the amount of data scanned during query execution.

5. Data Retrieval

The following SQL queries were implemented:

Query 1

Retrieve all purchases within a specified date range.

```
SELECT *  
FROM purchases  
WHERE purchase_date BETWEEN '2026-06-01' AND '2026-06-03';
```

Query 2

Retrieve all purchases belonging to a specified product category.

```
SELECT *  
FROM purchases  
WHERE product_category = 'Electronics';
```

The queries successfully returned the expected records.

6. Search Simulation

A Python program was developed to compare two searching techniques.

Sequential Search

Sequential Search examined each record one after another until the required Product ID was found.

Binary Search

Binary Search first sorted the records by Product ID before repeatedly dividing the search space into two halves until the required Product ID was located.

Execution time for both algorithms was measured using Python's `time.perf_counter()` function.

The results showed that **Binary Search executed faster than Sequential Search**, demonstrating the advantage of efficient searching techniques on sorted data.

7. Challenges Encountered

Several challenges were encountered during the implementation of the project.

- Incorrect SQL syntax during table creation.
 - CSV import failed because of an extra Timestamp column generated by Google Forms.
 - Date values in the CSV were stored in a format different from MySQL's DATE format.
 - These issues were resolved by correcting the SQL syntax, using temporary variables during data import, and converting the dates using SQL date conversion functions.
-

8. Conclusion

This project provided practical experience in data management, relational database design, indexing, SQL querying, and search algorithms.

The project demonstrated how structured data can be efficiently stored, retrieved, and analyzed using database technologies. It also showed the importance of indexing and efficient search algorithms in improving system performance when working with large datasets.